

Rainwater harvesting as a decentralized water conservation strategy for resource efficiency and green economy transition in Azerbaijan

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Keywords

green economy, rainwater harvesting, resource efficiency, water conservation, water security

ABSTRACT

Rainwater harvesting (RWH) offers a decentralized option for supplementing conventional water supplies while improving resource efficiency and urban resilience. This study integrates three independently prepared 2026 research projects on rainwater collection and use and places their core observations within a green-economy and policy framework for Azerbaijan. The source projects consistently described a simple system architecture based on roof or surface catchment, gutters and pipes, preliminary filtration, storage, distribution for non-potable uses and, where appropriate, overflow management or groundwater recharge. They emphasized irrigation, cleaning, car washing and toilet flushing as practical applications that can substitute treated freshwater. A structured qualitative synthesis was combined with targeted verification against recent scientific literature and Azerbaijan's water-policy documents. The analysis indicates that RWH can contribute to resource efficiency through freshwater substitution, local stormwater retention and diversification of water supply. Its economic value, however, is context dependent because installation, storage, pumping, maintenance, rainfall, water tariffs and end-use demand determine financial performance. Azerbaijan's National Strategy on Efficient Use of Water Resources for 2024–2040 explicitly calls for technical-economic assessment of rainwater collection and a pilot project for its use, strengthening the policy relevance of the approach. RWH should therefore be treated not as a stand-alone solution to water scarcity but as a complementary green-economy instrument that can reduce pressure on centralized water systems and freshwater sources when locally designed, safely operated and economically assessed.

INTRODUCTION

Water scarcity, climate variability and growing demand are increasing the need for diversified water-management strategies. Conventional water-supply systems depend on rivers, reservoirs, groundwater, treatment plants and extensive distribution networks, while many urban and peri-urban activities do not require drinking-water quality. Rainwater harvesting (RWH) addresses this mismatch by capturing precipitation close to where it falls and storing it for appropriate uses. In its simplest form, an RWH system contains a catchment surface, conveyance pipes, first-flush or filtration components, storage and a distribution mechanism. Modern systems can also be integrated with drainage, infiltration and building-water networks (Campisano et al., 2017).

The three source research projects that underpin this article approached RWH from a practical conservation perspective. Abbasli described rooftop capture, filtration, storage, non-potable uses and the possibility of directing excess water to recharge structures. Gurbanov emphasized the physical components of a collection system and the use of harvested water for irrigation and household or technical purposes. Azizli similarly focused on collection from roofs, storage in tanks and uses such as garden irrigation, cleaning and other technical activities. Across the three dossiers, the central idea is recognizable and consistent: using rainwater for suitable tasks can

reserve higher-quality freshwater for uses where it is genuinely required.

This conservation logic has a direct connection to the green economy. Resource efficiency seeks to generate greater social and economic value from a smaller quantity of natural resources and lower environmental pressure. RWH can support this objective by reducing demand for centrally treated water, moderating stormwater runoff and diversifying local supply. Reviews also show that the economic and environmental performance of RWH varies substantially by rainfall regime, tank size, demand profile, pumping energy, water tariffs and regulatory conditions; therefore, a system that is environmentally useful is not automatically financially attractive in every location (Campisano et al., 2017; de Sá Silva et al., 2022).

The objective of this study is to synthesize the three original research projects into an academic assessment of RWH as a decentralized water-conservation strategy for Azerbaijan. The paper asks three questions: (1) which technical and practical elements of RWH are consistently identified in the source projects; (2) through which pathways can RWH improve water-resource efficiency and green-economy resilience; and (3) what policy and economic conditions should be considered before wider implementation in Azerbaijan? The study does not estimate a national water-saving volume or financial payback because the source materials do not

contain measured rainfall series, roof areas, tank costs or water-demand records.

LITERATURE REVIEW

RWH has evolved from a traditional water-storage practice into a modern component of decentralized urban water management. Campisano et al. (2017) define urban RWH broadly as the collection, storage and treatment of runoff from rooftops and other impervious surfaces for on-site use. Common applications include toilet flushing, laundry, garden irrigation, cleaning and car washing. The review also highlights that RWH can provide multiple functions beyond water supply, including stormwater attenuation and, in some configurations, support for infiltration and local microclimate management.

Water quality is a decisive design consideration. The student projects correctly recognized the need for filtration before storage or use, but scientific evidence requires a more cautious distinction between potable and non-potable applications. Harvested rainwater quality depends on atmospheric deposition, roof material, antecedent dry periods, first-flush management, storage conditions and maintenance. Potable use therefore requires treatment appropriate to local health standards, whereas many non-potable uses can be supplied with simpler treatment and separation from drinking-water networks (Campisano et al., 2017). This distinction is important for scaling RWH safely.

The environmental and economic literature increasingly frames RWH as a multi-benefit system. De Sá Silva et al. (2022) identify decentralized water security, reduced potable-water demand, stormwater control and potential water-energy benefits as major advantages, while also noting that system economics can constrain uptake. Ali et al. (2025) similarly conclude that urban RWH can support sustainable development but that performance depends on rainfall, water demand, storage capacity, climate conditions and institutional support. These findings reinforce the need to evaluate RWH as an integrated infrastructure option rather than only as a household conservation device.

Financial feasibility is particularly context specific. Investment costs include tanks, filters, gutters, pumps, plumbing and installation, while operating costs may include energy, cleaning, inspection and component replacement. Benefits may include lower purchased-water consumption, reduced pressure on centralized infrastructure and avoided or reduced stormwater-management burdens. Jin et al. (2023) show that cost-benefit results are sensitive to system capacity and financial assumptions and that public support programs can materially affect feasibility. Consequently, green-economy arguments for RWH should not rely on universal payback claims; site-specific technical-economic assessment is required.

Azerbaijan's current policy environment makes this topic especially relevant. The National Strategy on Efficient Use of Water Resources for 2024–2040 identifies alternative water sources, wastewater reuse and rainwater management as elements of water security. Its action plan specifically calls for international experience to be reviewed, a technical-economic assessment of rainwater collection and use to be conducted and one pilot project to be implemented during the 2024–2027 phase (President of the Republic of Azerbaijan, 2024). In January 2026, a state program for Baku and the Absheron Peninsula also placed water losses, wastewater and stormwater infrastructure within a long-term urban water agenda (President of the Republic of Azerbaijan, 2026). This creates a clear policy bridge between the students' conservation ideas and national resource-efficiency priorities.

MATERIALS AND METHODS

The study used an integrative qualitative design based on three author-prepared research dossiers completed in 2026. Abbasli's English-language project used online research and scientific articles to explain rooftop RWH, filtration, storage, groundwater recharge and non-potable applications, with a specific interest in reducing pressure on freshwater resources such as the Kura River. Gurbanov's Azerbaijani-language project described rainwater characteristics, catchment surfaces, gutters, filtration, reservoirs and distribution and discussed examples of international use. Azizli's Azerbaijani-language project used online research, observation and visual sources to examine collection, storage, practical uses and conservation benefits.

The source materials were first coded into six analytical categories: catchment and conveyance; filtration and water quality; storage; end uses; freshwater-substitution potential; and economic or policy implications. Statements repeated across more than one project were treated as shared source findings. Claims requiring external support were then checked against peer-reviewed review studies and official Azerbaijani policy documents. External evidence is used to validate, qualify or extend the original projects rather than to replace their structure and central reasoning.

No measured rainfall, roof area, runoff coefficient, tank volume, water bill, installation cost or household demand dataset was supplied. Accordingly, no site-specific yield or payback is calculated. For future application, potential annual harvest can be estimated conceptually as $V = P \times A \times C$, where V is harvestable volume, P is rainfall depth, A is effective catchment area and C is a runoff coefficient representing losses. The equation is included as a sizing framework only; applying it requires local rainfall and building data. Economic assessment would additionally require capital and operating costs, water tariffs, discount rate and expected system life.

RESULTS AND DISCUSSION

System architecture identified in the source research

All three source projects describe a common RWH chain beginning with precipitation falling on a roof or other suitable surface. Gutters and downpipes convey the runoff toward a filter or first-flush component, after which water enters a storage tank. The stored water is then distributed by gravity or pumping for selected uses. Abbasli additionally highlighted diversion of surplus water to recharge structures, while Gurbanov emphasized the role of roof material and pre-storage filtration. Azizli's project presented the same system in a simplified conservation format, demonstrating that the basic technical logic was understood consistently across the independent works.

Figure 1 reorganizes these source-described elements into a single operational pathway. The figure does not represent a measured installation; it is a conceptual synthesis. Its main implication is that water quality and end use should be matched. Irrigation, outdoor cleaning, vehicle washing and toilet flushing are plausible non-potable applications, whereas any potable application would require substantially stronger treatment, monitoring and compliance with public-health requirements.

The design also has an urban drainage function. Storage creates temporary detention capacity, so part of a rainfall event is retained rather than immediately entering streets or drainage networks. The benefit depends on tank size, antecedent storage, rainfall intensity and how quickly stored water is used. Campisano et al. (2017) and Ali et al. (2025) both stress that multi-purpose performance should be incorporated into system design, because a tank optimized only for water supply may perform differently from one also intended for stormwater control.

Table 1 summarizes the main pathways through which the technical elements identified by the source projects can translate into environmental and green-economy value. These are qualitative pathways rather than monetized benefits.

Resource efficiency and pressure on freshwater systems

The strongest shared finding is the substitution principle: suitable rainwater can replace treated freshwater for activities that do not require potable quality. This is important because treatment and long-distance distribution consume infrastructure capacity, chemicals and energy. When a building or institution uses harvested water for irrigation or cleaning, the same service can be provided with less demand on centralized freshwater supply. De Sá Silva et al. (2022) note that this decentralization can support water conservation and, depending on the local system, may also reduce energy use associated with public supply.

For Azerbaijan, the relevance extends beyond household savings. Abbasli explicitly connected RWH to conservation of the Kura River and other freshwater sources. The present synthesis retains this logic but frames it cautiously: a single rooftop system cannot measurably change basin-scale abstraction, yet widespread substitution of non-potable demand can contribute to a portfolio of demand-management measures. The national strategy's inclusion of alternative sources and a rainwater pilot indicates that such distributed measures are being considered within a broader resource-efficiency approach (President of the Republic of Azerbaijan, 2024).

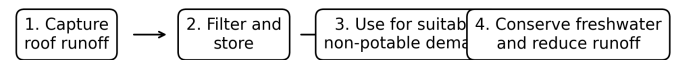
Figure 1 also highlights overflow and recharge as optional elements. Infiltration can be useful where soil, groundwater protection and engineering conditions permit it, but it should not be treated as universally appropriate. In dense urban areas, contaminated sites, high groundwater zones or poorly drained soils, controlled drainage may be preferable. This illustrates a broader point from the literature: RWH is decentralized, but it still requires site-specific engineering and water-quality management.

Table 1. Resource-efficiency and green-economy pathways of rainwater harvesting

RWH dimension	Resource-efficiency and economic pathway
Freshwater substitution	Reduces demand for treated freshwater and may lower water-service costs where tariffs and demand justify investment
Stormwater storage	Provides distributed detention and may reduce burden on urban drainage during suitable events
Irrigation support	Diversifies supply, protects treated water for higher-value uses and supports urban green infrastructure
Overflow and recharge	Can support local recharge and reduce runoff, subject to groundwater and soil safeguards
Financial feasibility	Site-specific economics depend on rainfall, demand, storage size, tariffs, energy use and possible incentives

Economic relevance and green economy transition

The economic case for RWH is broader than a household water bill. At user level, harvested rainwater can reduce consumption of metered mains water where tariffs and demand are sufficient to create savings. At system level, distributed storage can reduce peak demand on centralized supply and contribute to stormwater management. At environmental level, lower abstraction pressure can help protect natural capital embodied in rivers, reservoirs and groundwater. These benefits are consistent with green-economy principles because they seek to maintain services while using fewer treated freshwater resources and reducing pressure on public infrastructure.



Decentralized rainwater harvesting links local water supply with resource efficiency.

Figure 1. Conceptual rainwater harvesting pathway synthesized from the three source projects

Financial viability should nevertheless be stated carefully. The student dossiers describe RWH as inexpensive or capable of substantial savings, but they do not provide bills, equipment quotations or life-cycle costs that would permit those claims to be quantified. Published evidence shows mixed economic outcomes. Campisano et al. (2017) identify economic constraints as a major determinant of implementation, while Jin et al. (2023) demonstrate that tank sizing, inflation, discounting and public support can alter cost-benefit results. The economically appropriate question for Azerbaijan is therefore not whether RWH is always cheaper, but where it creates the greatest combined water-saving, drainage and resilience value per unit of investment.

Policy relevance and implementation priorities for Azerbaijan

The 2024–2040 National Strategy provides unusually direct policy support for this research topic. The action plan includes rainwater management and use, international benchmarking, technical-economic assessment and a pilot project. It also places alternative water sources within a broader program of infrastructure modernization and water-loss reduction (President of the Republic of Azerbaijan, 2024). This means that future school, municipal or building-scale demonstrations can be linked to a national policy agenda rather than presented only as environmental awareness activities.

Practical implementation framework and study limitations

Five priorities emerge from the combined evidence. First, RWH should initially target non-potable uses with predictable demand, such as irrigation, cleaning and toilet flushing. Second, roof and storage-water quality should be managed through first-flush diversion, filtration and regular maintenance. Third, tanks should be sized using local rainfall and demand data rather than generic assumptions. Fourth, economic appraisal should include capital cost, maintenance, pumping energy, water tariff savings and stormwater benefits where they can be valued. Fifth, pilot systems in schools or public buildings could combine water saving with environmental education while generating measured data for later scaling.

The study is limited by the nature of the underlying projects. They are educational research dossiers rather than engineering trials. No physical RWH system was monitored, no water-quality samples were analyzed and no measured economic dataset was available. Some numeric savings stated in the original projects could not be reconstructed from raw data and are therefore not repeated as findings. The green-economy discussion is qualitative and does not claim a specific payback period, national water-saving volume or reduction in Kura abstraction. Future work should instrument a pilot building, record rainfall and tank levels, measure non-potable demand and water quality, and conduct a full life-cycle cost assessment.

CONCLUSIONS

This study integrated three original research projects into an academic assessment of rainwater harvesting as a decentralized strategy for water conservation, resource efficiency and green economy transition in Azerbaijan. The source projects converge on a technically coherent

system comprising catchment, conveyance, filtration, storage and distribution, with irrigation, cleaning, vehicle washing and toilet flushing among the principal non-potable applications. Scientific literature supports the central conservation logic but also shows that performance depends on rainfall, demand, storage design, water quality, maintenance and economic conditions. The key green-economy contribution of RWH is its potential to substitute treated freshwater for suitable uses, diversify local supply and provide additional stormwater-management value. Azerbaijan's National Strategy on Efficient Use of Water Resources for 2024–2040 strengthens the relevance of this approach by explicitly requiring technical-economic assessment and a rainwater-use pilot project. RWH should therefore be developed as a complementary element of integrated water management rather than as a universal substitute for conventional supply. The next research step should move from descriptive assessment to monitored pilot systems that quantify water yield, quality, operating cost and avoided mains-water demand under Azerbaijani conditions.

AUTHOR CONTRIBUTIONS

T.A., H.G. and Z.A. contributed the original research materials, topic development, source collection and preliminary assessment of rainwater-harvesting methods and uses. A.M. contributed supervision of manuscript integration, methodological framing, green-economy interpretation, review and editing. All authors contributed to interpretation, reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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This research received no external funding.

DATA AVAILABILITY

This study is a qualitative synthesis of three author-prepared research dossiers and published literature. No primary hydrological, water-quality

or cost dataset was generated. The source-level descriptive materials supporting the synthesis are contained in the authors' original research files; no separate numerical dataset is available.

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